| EXP:1a | SETTING UP THE ENVIRONMENT AND PREPROCESSING THE DATA |
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**AIM:**

To preprocess the Titanic dataset by handling missing values, encoding categorical variables, performing feature scaling, and splitting the dataset into training and testing sets for machine learning applications.

**PROCEDURE:**

1. Import the required Python libraries such as Pandas, NumPy, Seaborn, and Scikit-learn.
2. Load the Titanic dataset using the Seaborn library.
3. Display the first few records and explore the dataset using information, statistical summary, and missing value analysis.
4. Handle missing values by filling the **Age** column with the median value and the **Embark Town** column with the mode value.
5. Remove the **Deck** column due to a large number of missing values.
6. Encode the categorical columns **Sex** and **Embark Town** using Label Encoding.
7. Remove unnecessary columns that are not required for model training.
8. Standardize the numerical features **Age** and **Fare** using StandardScaler.
9. Split the dataset into training and testing sets using the train-test split method.
10. Display the shape of the training and testing datasets.

**CODE:**

```
import pandas as pd
import seaborn as sns
from sklearn.model_selection import train_test_split
from sklearn.preprocessing import StandardScaler
df = sns.load_dataset("titanic")
print("Original Dataset:")
print(df.head())
print("\nMissing Values:")
print(df.isnull().sum())
df["age"] = df["age"].fillna(df["age"].median())
df["fare"] = df["fare"].fillna(df["fare"].median())
df["embark_town"] = df["embark_town"].fillna("Unknown")
df = df.drop(columns=["deck", "embarked"])
df["sex"] = df["sex"].map({"male": 0, "female": 1})
df["embark_town"] = df["embark_town"].astype("category").cat.codes
df = df.drop(columns=[
    "class", "who", "adult_male", "alive", "alone"])
numeric_columns = ["age", "fare", "sibsp", "parch"]
scaler = StandardScaler()
df[numeric_columns] = scaler.fit_transform(df[numeric_columns])
X = df.drop(columns=["survived"])
y = df["survived"]
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.25, random_state=10)
print("\nProcessed Dataset:")
print(df.head())
print("\nTraining Data Shape:", X_train.shape)
print("Testing Data Shape:", X_test.shape)
print("\nTraining Data:")
```

print(X_train.head())

## OUTPUT:

```
•••  Original Dataset:
       survived  pclass     sex   age  sibsp  parch     fare embarked  class  \
    0         0       3    male  22.0      1      0   7.2500        S  Third
    1         1       1  female  38.0      1      0  71.2833        C  First
    2         1       3  female  26.0      0      0   7.9250        S  Third
    3         1       1  female  35.0      1      0  53.1000        S  First
    4         0       3    male  35.0      0      0   8.0500        S  Third

         who  adult_male deck  embark_town alive  alone
    0    man        True  NaN  Southampton    no  False
    1  woman       False    C    Cherbourg   yes  False
    2  woman       False  NaN  Southampton   yes   True
    3  woman       False    C  Southampton   yes  False
    4    man        True  NaN  Southampton    no   True

    Missing Values:
    survived         0
    pclass           0
    sex              0
    age            177
    sibsp            0
    parch            0
    fare             0
    embarked         2
    class            0
    who              0
    adult_male       0
    deck           688
    embark_town      2
    alive            0
    alone            0
    dtype: int64

    Processed Dataset:
       survived  pclass  sex       age     sibsp     parch      fare  embark_town
    0         0       3    0 -0.565736  0.432793 -0.473674 -0.502445            2
    1         1       1    1  0.663861  0.432793 -0.473674  0.786845            0
    2         1       3    1 -0.258337 -0.474545 -0.473674 -0.488854            2
    3         1       1    1  0.433312  0.432793 -0.473674  0.420730            2
    4         0       3    0  0.433312 -0.474545 -0.473674 -0.486337            2

    Training Data Shape: (668, 7)
    Testing Data Shape: (223, 7)

    Training Data:
         pclass  sex       age     sibsp     parch      fare  embark_town
    578       3    1 -0.104637  0.432793 -0.473674 -0.357308            0
    323       2    1 -0.565736  0.432793  0.767630 -0.064516            2
    654       3    1 -0.873136 -0.474545 -0.473674 -0.512513            1
    157       3    0  0.049062 -0.474545 -0.473674 -0.486337            2
    692       3    0 -0.104637 -0.474545 -0.473674  0.489104            2
```

## RESULT:

The Titanic dataset was successfully preprocessed by handling missing values, encoding categorical variables, applying feature scaling, and splitting the dataset into training and testing sets. The processed data is now suitable for further machine learning model development.